LRT-Based Modification Indices and Exact Expected Parameter Change

June 25, 2026

Purpose

A modification index (MI) asks, for a single constrained parameter, how much fit would improve if the constraint were freed. In measurement-invariance work the constraint of interest is a cross-group equality (a loading or intercept tied across groups), and the MI localizes *which* item is non-invariant. magmaan ships the classical one-step form: a Lagrange-multiplier (score) statistic evaluated once at the constrained fit, with robust (sandwich-scaled) variants.

This note records the refit-based likelihood-ratio (LRT) counterpart, why it is the correct tool when the model is structurally misspecified elsewhere, and what it costs. The headline cost result is that the refit sweep is cheap: a full invariance sweep over a 20-item two-group model runs in about one tenth of a second, cold-started, in pure R.

1 The three localizations and their anchor points

Let H_0 (constrained) be nested in H_1 (the one extra constraint released), with m released degrees of freedom ($m = 1$ for a single equality). The same null can be probed from three places:

- **Score / LM (the MI).** Fit only H_0; test whether the score for the frozen direction is far from zero. One fit, anchored at the *constrained* model.
- **Wald.** Fit only H_1; test whether $R\hat{\theta}_1 = q$. One fit, anchored at the *free* model.
- **LRT.** Fit *both*; compare $T_d = T_0 - T_1 = 2N(F_0 - F_1)$.

All three converge to χ^2_m under the null and are asymptotically equivalent. They differ in finite samples and, decisively, in their robustness to misspecification *elsewhere in the model.*

2 Two senses of misspecification

Write the candidate as direction j and suppose the model is also wrong in some other direction k.

Distributional misspecification (non-normality, an inefficient weight $W \neq \Gamma^{-1}$, the failure of the information equality $\mathcal{I} = -\mathbb{E}[H]$). This inflates or deflates the *reference distribution* of the statistic. The robust score test repairs it by building the meat from $\hat{\Gamma}$ rather than assuming $\mathcal{I}$; the per-direction scaling $c = g^\top B_1 g / g^\top A_1 g$ restores a valid χ^2 under the local null. This is a *denominator* correction.

Structural misspecification (the model is wrong in direction $k \neq j$). The score $s_j(\hat{\theta}_0)$ is evaluated at the constrained solution, which the estimator has biased by spreading the k-misfit across every frozen parameter. This is a *numerator* contamination, and no sandwich touches it.

Proposition 1 (Why robustifying the MI cannot fix structural leakage)**.** *Under a fixed structural error in direction k, the contaminating part of $s_j(\hat{\theta}_0)$ is $O(\sqrt{N})$ (its noncentrality grows with N; that is how the test gains power against k), whereas any meat/bread reweighting is an $O(1)$ rescaling of the statistic. An $O(1)$ factor cannot recenter an $O(\sqrt{N})$ bias.*

The robust MI is therefore correctly sized under the local null ("j constrained, rest correct, distribution possibly wrong") and silent about structural bias. The score test's anchor model is the constrained one, which under genuine non-invariance *is* the misspecified model, so its per-item MIs inherit the leakage. The Wald's anchor (the free model) is, for a loadings-only alternative, correctly specified; this is the structural reason a Wald localization resists the contamination a score MI suffers.

3 The LRT modification index

To decontaminate the numerator one must stop conditioning on the suspect parameters at wrong fixed values. The LRT does this by *refitting*: for each candidate constraint, fit the released model H_1 and report

$$T_d = T_0 - T_1, \qquad \mathrm{df} = m,$$

together with the robust reference-law family from the Satorra (2000) difference spectrum. Let $\lambda_1, \ldots, \lambda_m$ be the eigenvalues of the difference operator; then

$$T_d \xrightarrow{d} \sum_{j=1}^{m} \lambda_j \chi^2_{1,j},$$

and magmaan reports the naive χ^2_m tail, the Satorra–Bentler mean scaling $\bar{c} = \frac{1}{m}\sum_j \lambda_j$, the mean–variance adjustment, the scaled-shifted statistic, the exact mixture tail (Imhof/Farebrother), and the penalized FMG/pEBA transform. None of these cost extra: they are functions of the same spectrum the nested difference test already computes.

Exact EPC. The one-step expected parameter change is a single Newton step, epc $= s_{\mathrm{eff}}/\mathcal{I}_{\mathrm{eff}}$. The LRT counterpart is the *refitted* value of the freed parameter minus its constrained value: no linearization, the actual change. The existing standardization (`epc.lv`, `epc.all`) applies unchanged.

4 Conditional localization: exact, not local

Because the LRT names both endpoints, it can *condition.* To test item j while protecting against a suspected non-invariant set B, free B in *both* H_0 and H_1 (in magmaan, `group_partial`). Then B's misfit is identical on the two sides and cancels in T_d, so j is isolated.

This is the refit analogue of the Bera–Yoon (1993) adjusted score test, which orthogonalizes s_j against the k-direction. The crucial difference: Bera–Yoon removes the *linear* dependence and is valid only under local ($O(N^{-1/2})$) misspecification; the conditional LRT removes B's misfit to all orders by genuinely refitting, so it is exact under fixed non-invariance. The two extremes of the conditioning dial are the ordinary MI (condition on everything, maximal contamination) and the configural Wald (condition on nothing, zero contamination, all df spent).

Remark. The residual cost is the specification-search/anchor-item problem: a third non-invariant item left in the tied background contaminates both the j-test and the choice of B. And the difference spectrum is evaluated at H_1's pseudo-true point; the exact mixture fixes the reference-law *shape*, while conditioning fixes the *point* at which it is evaluated. These are orthogonal jobs.

5 Cost

The refit sweep is #candidates nested fits plus one difference spectrum each, against the one fit of the score sweep. Table 1 times both for a two-group one-factor CFA (loadings 0.6, one planted non-invariant loading, $n_A : n_B = 1{:}3$), cold-started (magmaan computes its own start values).

Table 1: Cold-start LRT-MI vs. one-step score sweep, $n = 1000$, median of 3 runs.

items	npar	cand	one-step (ms)	LRT (ms)	per cand (ms)	LRT/one-step
6	24	5	1.0	6.0	1.2	6.0
9	36	8	1.0	15.0	1.9	15.0
12	48	11	1.0	23.0	2.1	23.0
15	60	14	2.0	43.0	3.1	21.5
20	80	19	6.0	97.0	5.1	16.2

The ratio LRT/one-step is 6–23×, but only because the one-step sweep is near-free; in absolute terms the entire LRT sweep is sub-100 ms even at 20 items. Sample size enters mildly through the empirical $\hat{\Gamma}$: at 12 items the per-candidate cost grows from 2.0 ms at $n = 500$ to 4.7 ms at $n = 8000$. The conclusion for the design question is unambiguous: at the model sizes measurement-invariance studies use, the refit cost is not a barrier.

Implementation. The sweep ships as two R functions, mirroring the one-step pair (`modification_indices` for absent parameters, `score_tests` for equality releases), each a thin composition of the one-call estimator `magmaan` and the existing nested tests. `modification_indices_lrt(fit, data)` is the refit of the lavaan `modindices` table: for every parameter absent from the model (cross loadings, residual covariances) it refits the augmented model and reports the nested $\Delta\chi^2$ and the exact refitted change next to the one-step `mi`/`epc`. Each row also carries an observed-bread robust p-value `lrt_p_obs`: the exact eigenvalue-mixture tail of the observed-Hessian profile-LRT (`ml_profile_lrt`), which is the *model-misspecification*-robust reference law for the difference statistic (the bread is the profile Hessian Q, not the expected information). It is the "observed" axis, distinct from the empirical-Γ Satorra–Bentler reference, which is the *non-normality* axis. The profile contrast lives in the first-stage moment space, so adding a parameter (which changes the parameter count) needs no restriction-map pinning, and its spectrum size exceeds the nominal 1 df; the mixture tail, not the mean-scaled statistic, is the well-behaved summary. The observed-bread column dispatches on the anchor's estimator: complete-data normal-theory ML uses `ml_profile_lrt`; all-ordinal and mixed-ordinal DWLS fits use the categorical estimated-weight profile-LRT (`ordinal_profile_lrt`, `mixed_ordinal_profile_lrt`) over the polychoric NACOV; continuous ULS/GLS use `continuous_ls_profile_lrt` (one shared weight, empirical Γ from the raw data); and FIML and two-stage ML (ML2S) use `fiml_profile_lrt` / `two_stage_nt_profile_lrt`. For the missing-data estimators the plain $\Delta\chi^2$ cannot use $2N \cdot f_{\min}$ (the model-vs-saturated FIML chi-square is not $2N \cdot f_{\min}$, and ML2S works on the Stage-1 EM moments), so the same profile-LRT call supplies the plain difference (`T_diff`/`p_unscaled`, where the saturated term cancels) along with the robust mixture. Continuous WLS is supported too, but since its weight is not retained on the fit the same W must be passed via `modification_indices_lrt(..., weight=W)` (it is threaded through both the augmented refit and the profile-LRT). The plain $\Delta\chi^2$ columns still report for any complete-data estimator. `score_tests_lrt(fit, data)` is the equality-release sweep: for each cross-group tied family it refits the released model and runs the Satorra-2000 difference test, returning the reference-law family and the exact released per-group estimates; an existing `group_partial` baseline on the anchor is carried into every refit, so the conditional localization is automatic. For both complete-data ML and FIML the equality-release statistic reproduces a hand-built two-fit `robust_nested_lrt` to machine precision.

6 The decision layer: corrected significance and substantive effect

The per-candidate columns are raw. Turning them into a usable modification index needs two *orthogonal* corrections plus an effect-size gate.

Two orthogonal corrections. The observed-bread reference law `lrt_p_obs` fixes the *per-test* size under model misspecification (Section 2); it is not a multiplicity fix. Testing K candidates inflates the *family-wise* error on top of that, which a Benjamini–Hochberg adjustment of the robust p controls (`lrt_p_obs_bh`); FDR is preferred to Bonferroni for the large candidate sets a modindex sweep produces (Thissen, Steinberg & Kuang, 2002). The two act on different failures and compose: apply BH to the robust p, never to the naive one.

Effect size. Significance is necessary but not sufficient: in large N a negligible misspecification is highly significant, and under collinearity a real one can be non-significant. The companion axis is the *exact* refit EPC standardized to `std.all` (`sepc_lrt`), so one correlation-scale threshold $|\cdot| \geq \tau$ applies uniformly across loadings and covariances.

The verdict. Combining corrected significance with substantive size is the Saris–Satorra–Sörbom 2×2, here upgraded on every axis (refit LRT not score MI; observed-bread not naive χ^2; FDR not nothing; exact not one-step EPC):

	$\lvert\text{sepc}\rvert \geq \tau$	$\lvert\text{sepc}\rvert < \tau$
$p_{\text{bh}} \leq \alpha$	**free** (real release)	**trivial** (large-N artifact)
$p_{\text{bh}} > \alpha$	**underpowered** (flag, hold)	**ok** (leave fixed)

This ships as `mi_screen(table, alpha, effect_min, adjust)`, a composable layer that screens either the `modification_indices_lrt` or the `score_tests_lrt` table (it auto-resolves the robust p and effect columns). None of it makes the search confirmatory: a data-driven freed parameter is a hypothesis, and the residual safeguard is replication on held-out data.

What remains as an optimization. This baseline cold-starts every released fit. Starting each released fit from the constrained MLE (the C++ core takes an explicit `x0` and contracts it into the released parameter space) should cut the per-candidate refit to a few Newton steps. The benchmark (`benchmarks/r/bench_mi_lrt.R`) is structured to quantify that warm-start speed-up against this cold-start reference once the warm path lands.